Задание 1 (a)

10 июля 2026 г. 11:38

$$\lim_{x\to 0}\frac{1-\cos 6x}{x^2}=\lim_{x\to 0}\frac{2\sin^2 3x}{x^2}=$$

$$\lim_{x\to 0}\frac{\sin x}{x}\to 1$$

$$\frac{0}{0}$$

$$=2\cdot\lim_{x\to 0}\frac{9\sin^2 3x}{9x^2}=18\cdot\lim_{x\to 0}\left(\frac{\sin 3x}{3x}\right)^2\to 18$$

$$\frac{\sin 3x}{3x}\to 1$$

Задание 1 (б)

10 июля 2026 г. 12:26

$$\lim_{x\to\infty} x\left(\sqrt{x^2+1}-x\right) = \lim_{x\to\infty} x\sqrt{x^2+1}-x^2 =$$

$$=\lim_{x\to\infty}\sqrt{x^4+x^2}-x^2 = \lim_{x\to\infty}\frac{\left(\sqrt{x^4+x^2}-x^2\right)\left(\sqrt{x^4+x^2}+x^2\right)}{\sqrt{x^4+x^2}+x^2} =$$

$$=\lim_{x\to\infty}\frac{\left(\sqrt{x^4+x^2}\right)^2-\left(x^2\right)^2}{\sqrt{x^4+x^2}+x^2} = \lim_{x\to\infty}\frac{x^2}{\sqrt{x^4+x^2}+x^2} \left(\cdot\frac{\frac{1}{x^2}}{\frac{1}{x^2}}\right) =$$

$$=\lim_{x\to\infty}\frac{1}{\frac{1}{x^2}\sqrt{x^4+x^2}+1} = \lim_{x\to\infty}\frac{1}{\sqrt{1+\underbrace{\frac{1}{x^2}}_{0}}+1} \to \frac{1}{2}$$

$$\int_5^{12} \frac{x}{\sqrt{x+4}}\,dx =$$

$$t = x+4, \quad x = t-4$$

$$dt = d(x+4) = (x+4)'dx = dx$$

$$\int \frac{t-4}{\sqrt{t}}\,dt = \int \frac{t^1}{t^{\frac{1}{2}}}\,dt - 4\int t^{-\frac{1}{2}}\,dt =$$

$$= \frac{t^{\frac{1}{2}+1}}{\frac{1}{2}+1} - 4\cdot\frac{t^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} = \frac{2t^{\frac{3}{2}}}{3} - \frac{8t^{\frac{1}{2}}}{1} = \frac{2t^{\frac{3}{2}} - 24t^{\frac{1}{2}}}{3} =$$

$$= \frac{2(x+4)^{\frac{3}{2}} - 24(x+4)^{\frac{1}{2}}}{3} = \frac{2(x+4)\sqrt{x+4} - 24\sqrt{x+4}}{3} =$$

$$= \frac{\sqrt{x+4}\,(2x-16)}{3} = \frac{2(x-8)\sqrt{x+4}}{3} + C$$

$$= \frac{2(x-8)\sqrt{x+4}}{3}\Bigg|_5^{12} = \frac{2(12-8)\sqrt{12+4} - 2(5-8)\sqrt{5+4}}{3} =$$

$$= \frac{2\cdot4\cdot4 - 2(-3)\cdot3}{3} = \frac{32+18}{3} = \frac{50}{3}$$

Задание 4

12 июля 2026 г. 16:11

$$\begin{vmatrix} -2 & 1 & 2 & 1 \\ 2 & 0 & 4 & 3 \\ -2 & 1 & 1 & 1 \\ -1 & 2 & -3 & 1 \end{vmatrix} \times(-1) = \begin{vmatrix} -2 & 0 & 2 & 1 \\ 2 & -3 & 4 & 3 \\ -2 & 0 & 1 & 1 \\ -1 & 1 & -3 & 1 \end{vmatrix} = -3\begin{vmatrix} -2 & 2 & 1 \\ -2 & 1 & 1 \\ -1 & -3 & 1 \end{vmatrix} + \begin{vmatrix} -2 & 2 & 1 \\ 2 & 4 & 3 \\ -2 & 1 & 1 \end{vmatrix} =$$

$$= -3\left(\begin{vmatrix} -2 & 1 \\ -1 & -3 \end{vmatrix} - \begin{vmatrix} -2 & 2 \\ -1 & -3 \end{vmatrix} + \begin{vmatrix} -2 & 2 \\ -2 & 1 \end{vmatrix} \right) + \left(\begin{vmatrix} 2 & 4 \\ -2 & 1 \end{vmatrix} - 3\begin{vmatrix} -2 & 2 \\ -2 & 1 \end{vmatrix} + \begin{vmatrix} -2 & 2 \\ 2 & 4 \end{vmatrix} \right) =$$

$$= -3(6+1-(6+2)+(-2+4)) + (2+8-3(-2+4)+(-8-4)) =$$

$$= -3(7-8+2)+(10-6-12) =$$

$$= -3-8 = -11$$